Routing Table

A Routing Table is a database maintained by a router or host that stores information about available network routes. It helps the router determine the best path for forwarding packets from the source to the destination network.

Each time a packet arrives, the router checks the routing table to decide where the packet should be sent next.

Components of a Routing Table

A routing table typically contains the following fields:

1. **Destination Network**

Specifies the target network address where the packet needs to go.

2. **Subnet Mask / Prefix Length**

Defines the size of the destination network and helps the router identify the correct network range.

3. **Next Hop**

Indicates the IP address of the next router to which the packet should be forwarded.

4. **Outgoing Interface**

Specifies the interface through which the packet will leave the router.

5. **Metric**

Represents the cost of a route. Lower metric values indicate better or more preferred routes.

Packet Forwarding Mechanism

Packet forwarding is the process where a router receives data on one interface and sends it out another interface toward its destination. It relies on a routing table, which acts as a router's roadmap, matching destination IP addresses to specific "next hops" or physical exit interfaces.

Packet Forwarding Process in a Router

1. Ingress and Decapsulation

When a packet arrives at a router through an incoming interface, the router first removes the Layer 2 header and trailer. This process is called **decapsulation**. After removing the Layer 2 information, the router gains access to the Layer 3 packet, which is usually an IP packet.

2. Routing Table Lookup

The router examines the **Destination IP Address** contained in the IP packet header. It then compares this address with the entries stored in its routing table to determine the best possible path for forwarding the packet.

Longest Prefix Match

If multiple routes match the destination IP address, the router selects the route with the **most specific match**, meaning the route with the **longest subnet mask (prefix length)**.

3. Route Evaluation Components

Each routing table entry contains important information that guides the forwarding process:

- **Network ID / Subnet Mask:**
  Defines the destination network range.
- **Next Hop:**
  Specifies the IP address of the next router along the path. If the destination belongs to a directly connected network, the next hop is the destination host itself.
- **Outgoing (Exit) Interface:**
  Indicates the physical or logical interface through which the packet must be forwarded.
- **Metric:**
  Represents the cost of a route. When multiple routes to the same destination exist, the route with the lowest metric is preferred.

4. Packet Modification

Before forwarding the packet, the router performs certain modifications:

- The **TTL (Time-to-Live)** value in the IP header is decreased by 1. This prevents packets from circulating endlessly in the network.
- Since the TTL value changes, the router recalculates the **IP Header Checksum** to maintain packet integrity.

5. Encapsulation and Egress

The router then prepares the packet for transmission on the next network segment.

- It creates a new Layer 2 frame appropriate for the outgoing network.
- The router determines the destination MAC address of the next hop by consulting the **ARP (Address Resolution Protocol) table**.
- Finally, the IP packet is encapsulated into the new frame and transmitted through the designated outgoing interface.

This complete process is repeated at every router along the communication path until the packet successfully reaches its final destination.

IPv4 Packet Format:

<table>
<tr><td>Version
(8 bits)</td><td>Header
Length
(8 bits)</td><td>Type of Service
(8 bits)</td><td colspan="2">Total Length
(8 bits)</td></tr>
<tr><td colspan="3">Identification
(16 bits)</td><td>Flags
(3 bits)</td><td>Fragment Offset
(13 bits)</td></tr>
<tr><td colspan="2">Time to Live (TTL)
(8 bits)</td><td>Protocol
(8 bits)</td><td colspan="2">Header Checksum
(16 bits)</td></tr>
<tr><td colspan="5">Source IP Address
(32 bits)</td></tr>
<tr><td colspan="5">Destination IP Address
(32 bits)</td></tr>
<tr><td colspan="5">Options</td></tr>
<tr><td colspan="5">Data</td></tr>
</table>

Version (4 bits): This field identifies the version of IP being used. For IPv4, this is always 4 (binary 0100).

Internet Header Length (4 bits): Specifies the size of the header in **32-bit words**.

Header Length=IHL×4

Minimum value of IHL is 5 (5X4=20 bytes). If Options is present, it can go up to 15 (15 X4=60 bytes).

Total Length (16 bits): Specifies total packet size

Maximum possible value:

$$2^{16} - 1 \\ = 65535\ bytes$$

Example

Header:20 bytes

Data: 1000 bytes

Thus:

Total Length = 1020 bytes

<u>Identification (16 bits):</u> If IP packet is fragmented during the transmission, all the fragments contain same identification number to identify original IP packet they belong to.

If Identification of large packet is 2500

Suppose a large packet is divided into 3 fragments: Fragment 1, Fragment 2, Fragment 3

Then, Identification of Fragment 1= 2500

Identification of Fragment 2= 2500

Identification of Fragment 3= 2500

<u>Flags (3 bits):</u> if IP Packet is too large to handle, these flags tells if they can be fragmented or not. In this 3-bit flag, the MSB is always set to 0.

0	**DF**	**MF**

If DF=1, it means fragmentation is not required.

MF=1, it means more fragmentation is required.

Example:

Suppose a sender wants to transmit a packet with **4,000 bytes** of data over a network with an MTU of **1,500 bytes**.
Because the standard IPv4 header is **20 bytes**, the maximum data size per packet is =(1500 – 20) bytes = 1480 bytes. The data must be split into three fragments.

Fragment 1

- **Data Size:** 1,480 bytes
- **Flags:** DF = 0, **MF = 1** (More fragments are coming)

Fragment 2

- **Data Size:** 1,480 bytes
- **Flags:** DF = 0, **MF = 1** (More fragments are coming)

Fragment 3

- **Data Size:** 1,040 bytes \((4000 - 1480 - 1480 = 1040)\)
- **Flags:** DF = 0, **MF = 0** (This is the final fragment)

<u>Fragment Offset (13 bits):</u> Specifies position of a fragment within original packet.

Suppose a sender wants to transmit a packet with **4,000 bytes** of data over a network with an MTU of **1,500 bytes**.

Because the standard IPv4 header is **20 bytes**, the maximum data size per packet is \(1500 - 20 = 1480) bytes. The data must be split into three fragments. [1, 2]

Fragment 1

- **Data Size:** 1,480 bytes
- **Flags:** DF = 0, **MF = 1** (More fragments are coming)
- **Fragment Offset:**0/8=0

Fragment 2

- **Data Size:** 1,480 bytes
- **Flags:** DF = 0, **MF = 1** (More fragments are coming)
- **Fragment Offset: (1480/8) =185**

Fragment 3

- **Data Size:** 1,040 bytes [4000 - 1480 - 1480 = 1040]
- **Flags:** DF = 0, **MF = 0** (This is the final fragment)
- **Fragment Offset:** (1480 + 1480)/ 8 = 2960/8 = 370

TTL (8 bits): To avoid looping in the network, every packet is sent with some TTL value set, which tells the network how many routers (hops) this packet can cross. At each hop, its value is decremented by one and when the value reaches zero, the packet is discarded.

If **TTL > 0** → packet continues forwarding

If **TTL = 0** → packet is discarded and an **ICMP Time Exceeded** message is sent back to the source

Example:

Suppose a computer with IP **192.168.1.10** sends a packet to **172.16.1.20**

Initial packet:

Source IP = 192.168.1.10
Destination IP = 172.16.1.20
TTL = 5

The packet passes through routers:

Router	**TTL Before**	**TTL After**
Source Computer	5	5
Router 1	5	4

Router	TTL Before	TTL After
Router 2	4	3
Router 3	3	2
Router 4	2	1
Router 5	1	0

At **Router 5**:

TTL = 0

The router discards the packet and sends message “Time Exceeded“ back to the source computer.

Protocol (8 bits): The **Protocol field** in the IPv4 header is an **8-bit field** that identifies **which upper-layer protocol (Transport/Network layer protocol)** should receive the data after the IP packet reaches the destination.

It tells the receiver **what type of data is carried in the packet**, such as TCP, UDP, ICMP, etc.

The receiver checks this field and forwards the payload to the correct protocol.

Common Protocol Numbers

Protocol	Decimal Value	Binary Value
ICMP	1	00000001
TCP	6	00000110
UDP	17	00010001

Example 1: Web Browsing using TCP

Suppose a user opens a website:

www.example.com

The browser sends an HTTP request using **TCP**.

IPv4 packet:

Version = 4
Source IP = 192.168.1.10
Destination IP = 142.250.190.78

TTL = 64
Protocol = 6
Data = HTTP Request

Here:

Protocol = 6

Meaning:

- The payload contains **TCP data**
- At the destination, the IP layer sends the data to **TCP**

Flow:

User → IP Packet → Protocol=6 → TCP → Web Server

Example 2: Video Streaming using UDP

Suppose a user watches a live video stream.

Packet:

Source IP = 192.168.1.10
Destination IP = 224.10.10.10
Protocol = 17
Data = Video Stream

Here:

Protocol = 17

Meaning:

- Payload contains **UDP data**
- Destination forwards data to **UDP**

Example 3: Ping Command using ICMP

Suppose you run:

ping google.com

Packet:

Source IP = 192.168.1.10
Destination IP = 142.250.190.78
Protocol = 1
Data = Echo Request

Here:

Protocol = 1

Meaning:

- Packet contains **ICMP data**
- Used for testing connectivity

Header Checksum (16 bits): This field is used to keep checksum value of entire header which is then used to check if the packet is received error-free

Source address (32 bits):32-bit address of the Sender (or source) of the packet.

Destination Address (32 bits): 32-bit address of the Receiver (or destination) of the packet.

Options: This is optional field, which is used if the value of IHL is greater than 5. These options may contain values for options such as Security, Record Route, Time Stamp, etc.

ARP (Address Resolution Protocol)

ARP (Address Resolution Protocol) maps an IPv4 address to the corresponding MAC address on a local network so frames can be delivered to the correct device.

- Resolves IP addresses to MAC addresses for local communication
- Works only within the same broadcast domain (LAN)
- Operates directly over link-layer technologies such as Ethernet
- Essential for delivering frames to the correct device

Important Components

ARP Cache

- A temporary storage table maintained by a device.
- Stores recently resolved IP-to-MAC address mappings.
- Reduces network overhead by avoiding repeated ARP requests.

ARP Cache Timeout

- The time duration for which an entry remains valid in the ARP cache.
- After expiration, the entry is removed to prevent the use of outdated mappings.

ARP Request

when the MAC address for a given IP is not present in the ARP cache. A broadcast message sent to all devices in the local network and asks: “Who has this IP address?”

ARP Reply (ARP Response)

- A unicast message sent by the device that owns the requested IP address containing the corresponding MAC address.
- Allows the sender to update its ARP cache and proceed with data transmission.

How ARP Works

The working of ARP can be explained in the following steps:

1. ARP Cache Lookup

The sender first checks its ARP cache for an existing entry corresponding to the destination IP address.

If a valid entry is found, the MAC address is used immediately and data transmission begins.

2. ARP Request Broadcast

If no cache entry exists, the sender generates an ARP request. This request is broadcast to all devices on the local network.

3. Request Processing by Hosts

Every device on the LAN receives the ARP request. Each device compares the requested IP address with its own IP address.

4. ARP Reply Transmission

The device whose IP address matches the request sends an ARP reply.The reply is sent as a unicast message and contains the device's MAC address.

5. ARP Cache Update

Upon receiving the reply, the sender stores the IP–MAC mapping in its ARP cache.

This cached entry is used for future communication until it expires.

Example 1:

If PC A wants to send data to PC B:

PC A checks whether PC B is in the same network.

If PC B is in the same network:

PC A sends an ARP request:

"Who has IP 192.168.1.10?"

The switch broadcasts that ARP request to all devices in the local network.

PC B replies with its MAC address.

Example 2:

Suppose PC A wants to send data to PC B:

If PC B is in another network:

PC A uses ARP to get the router's MAC address (default gateway).

The switch sends the frame to the router.

The router reads the destination IP and forwards the packet to the next network.

RARP (Reverse Address Resolution Protocol)

RARP (Reverse Address Resolution Protocol) is a network protocol that allows a device to discover its IP address when only its MAC (Media Access Control) address is known. It was designed for systems such as diskless workstations, which do not have permanent storage to save their IP addresses. These devices boot from ROM and must request their IP address dynamically from a RARP server on the local network.

Important Components of RARP

IP Address Assignment: Normally, a machine stores its IP address in a configuration file. Diskless systems cannot do this and rely on RARP for IP assignment.

Physical Address: Every network device has a unique MAC address, stored in its Network Interface Card (NIC).

RARP Request: A device broadcasts a request containing its MAC address to ask for the corresponding IP address.

RARP Server: The server maintains a mapping of MAC addresses to IP addresses. On receiving a request, it replies with the correct IP address.

Working of RARP

Reverse ARP (RARP) is a network protocol used by a client machine in a local area network (LAN) to obtain its Internet Protocol (IP) address from the gateway router's ARP (Address Resolution Protocol) table. When a machine doesn't have the memory to store its IP address, such as diskless machines or newly configured systems, it uses RARP to request an IP address.

1. **RARP Request**: A client broadcasts a RARP request containing its MAC address.

2. **Server Lookup**: A RARP server (or gateway router with ARP table) checks its mapping of MAC -> IP.

3. **RARP Reply**: If a match is found, the server responds with the client’s IP address.

4. **Client Configuration**: The client configures itself with the provided IP and can now communicate on the network.

Is RARP Obsolete?

Yes. RARP is obsolete and rarely used in modern networks because of its limitations. It has been replaced by:

1. **BOOTP (Bootstrap Protocol):** Provides IP along with gateway, DNS and configuration.

2. **DHCP (Dynamic Host Configuration Protocol):** Extension of BOOTP, most widely used today for dynamic IP assignment and full configuration support.

Issues in RARP

• Limited scalability

• Lack of security

• No subnetting support

• Router incompatibility

• Compatibility issues with modern networks

Difference between RARP and ARP

RARP	ARP
A protocol used to map a physical (MAC) address to an IP address	A protocol used to map an IP address to a physical (MAC) address
To obtain the IP address of a network device when only its MAC address is known	To obtain the MAC address of a network device when only its IP address is known
Client broadcasts its MAC address and requests an IP address and the server responds with the corresponding IP address	Client broadcasts its IP address and requests a MAC address and the server responds with the corresponding MAC address
Rarely used in modern networks as most devices have a pre-assigned IP address	Widely used in modern networks to resolve IP addresses to MAC addresses
RFC 903 Standardization	RFC 826 Standardization
It uses the value 3 for requests and 4 for responses	It uses the value 1 for requests and 2 for responses

ICMP

In networking, ICMP stands for Internet Control Message Protocol**.** Internet Control Message Protocol (ICMP) is a network layer protocol widely used by network devices such as routers, gateways and hosts to send error messages and operational information. Since the Internet Protocol (IP) itself does not have an inbuilt error-reporting or correction mechanism, ICMP is a supporting protocol within the IP suite that helps in reporting errors and sending diagnostic messages. ICMP is mainly used for:

- **Reporting delivery problems**

  If a message cannot be delivered, ICMP informs the source about the failure.
  Example: If a packet is too large and cannot be forwarded, the receiver drops the packet and sends an ICMP error message to the sender.

- **Diagnosing network connectivity**

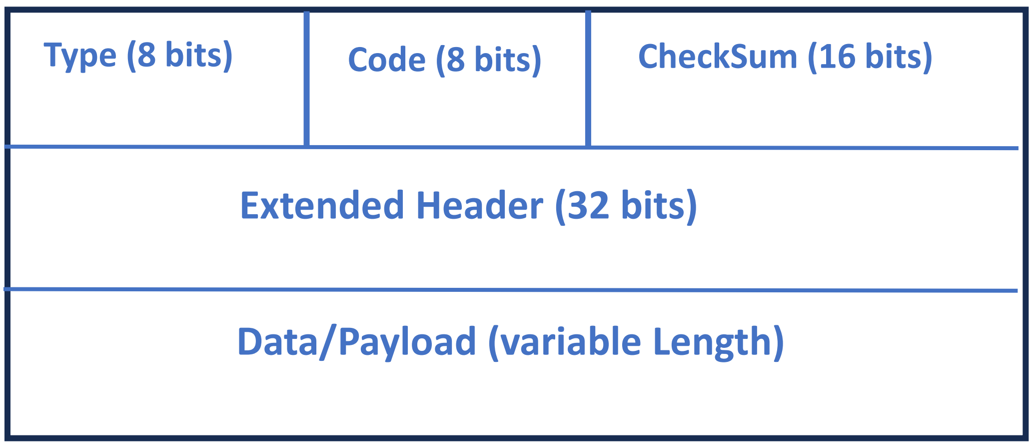
  Traceroute: Used to determine the path that packets take across routers to reach the destination.

  Ping: Sends echo request and echo reply messages to measure round-trip time and test network connectivity.

- **Providing status information between routers and hosts.**

Without ICMP, devices would not be able to inform the sender about delivery failures, congestion or misrouting, making network troubleshooting almost impossible.

ICMP Packet Format:



Type (8 bits)

- Identifies the **kind of ICMP message**
- Examples:
  - 8 → Echo Request (Ping)
  - 0 → Echo Reply

    - 3 → Destination Unreachable
    - 11 → Time Exceeded

2. Code (8 bits)

- Provides **additional details** about the Type
- Example:
    - Type 3 (Destination Unreachable)
        - Code 0 → Network unreachable
        - Code 1 → Host unreachable

3. Checksum (16 bits)

- Used for **error detection**
- Covers the entire ICMP message (header + data)

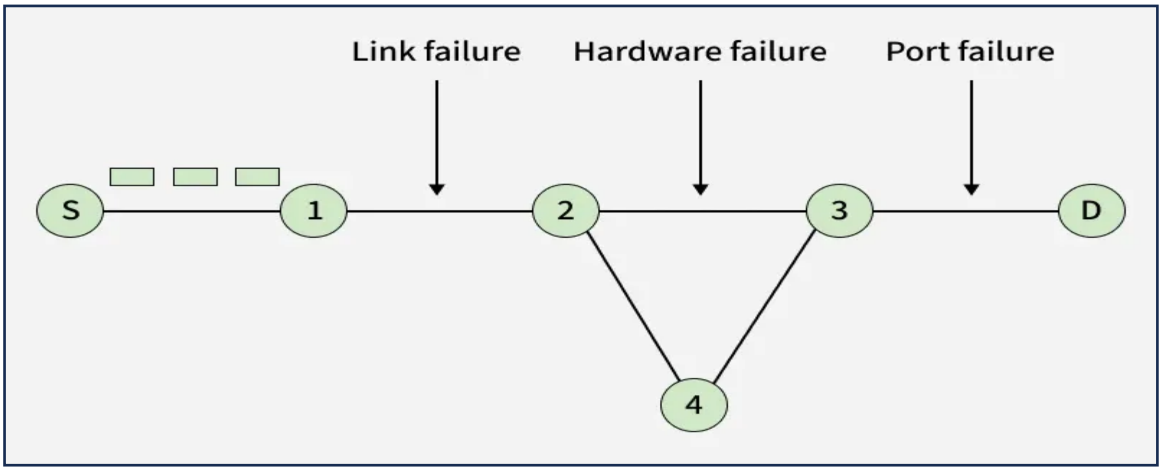
4. Rest of Header (32 bits)

- Depends on the **Type and Code**
- Examples:
    - For Echo Request/Reply:
        - Identifier (16 bits)
        - Sequence Number (16 bits)
    - For error messages:
        - Contains unused or specific control info

5. Data / Payload (Variable)

- Contains additional information
- For error messages:
    - Includes **original IP header + first 8 bytes of the original data**
- For echo (ping):
    - Contains arbitrary data (used to measure round-trip time)

1. Destination Unreachable:



Generated when a router or destination host cannot deliver the packet.

Type = 3

Common Codes

Code	Meaning
0	Network unreachable
1	Host unreachable
2	Protocol unreachable
3	Port unreachable
4	Fragmentation needed
5	Source route failed

Working

Example:

A sender sends data to another host.

- Router cannot find route
- OR destination host is down
- OR requested port is closed

Then router/host sends:

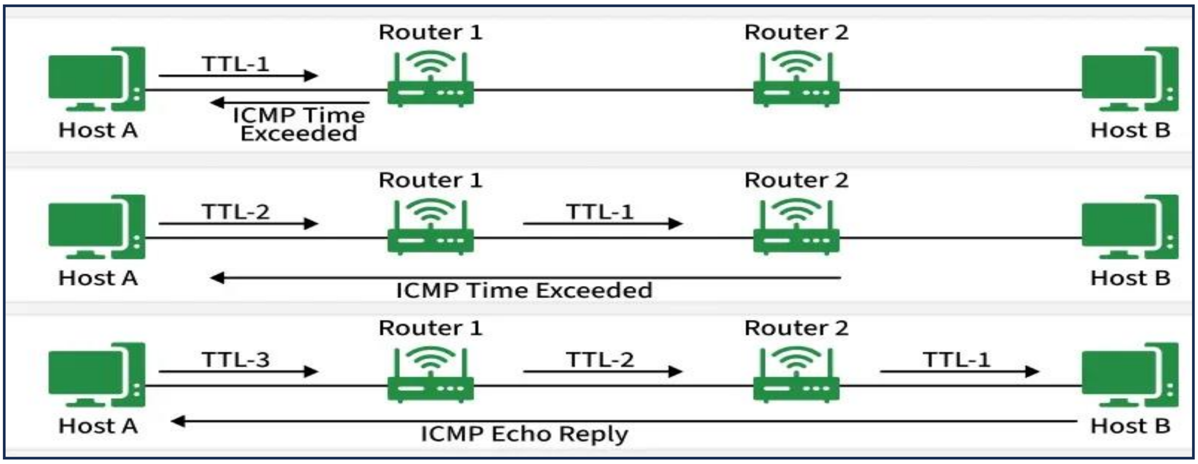
- Type = 3 with Appropriate code

Practical Example

When pinging a non-existing host:

"Destination Host Unreachable"

2. Time Exceeded Message



Generated when packet lifetime expires.

Type

Type = 11

Codes

Code	**Meaning**
0	TTL exceeded
1	Fragment reassembly time exceeded

Working

Each router decreases the **TTL (Time To Live)** value by 1.

When TTL becomes 0:

- Router discards packet
- Sends ICMP Time Exceeded message

Used In

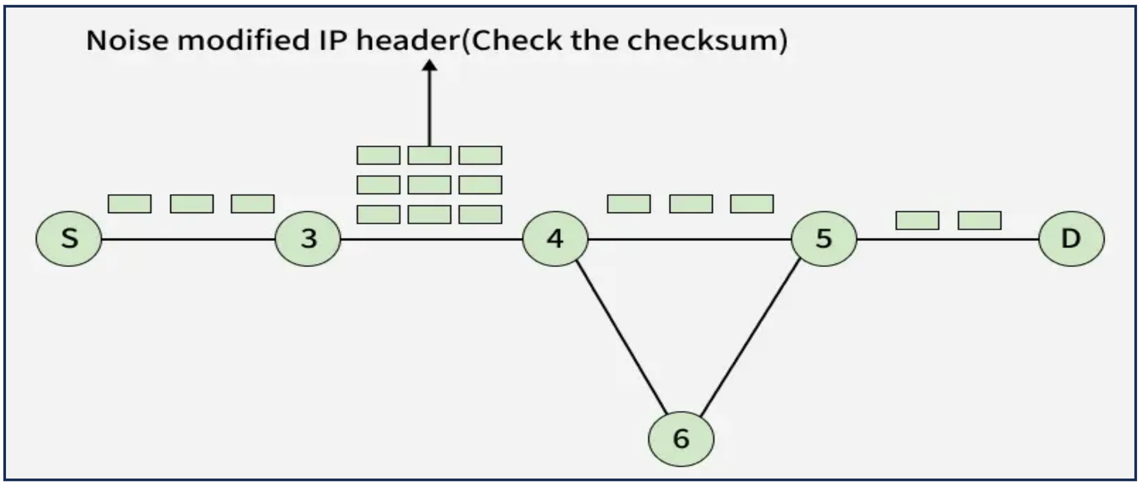
Traceroute Command

Traceroute intentionally sends packets with small TTL values to discover routers in path.

Example

TTL = 1
First router decreases TTL to 0 → packet discarded → ICMP sent back.

3. Parameter Problem Message



Generated when router detects an invalid value in IP header.

Type

Type = 12

Causes

- Incorrect header field
- Invalid option
- Missing required information

Working

Router checks IP header while forwarding packet.

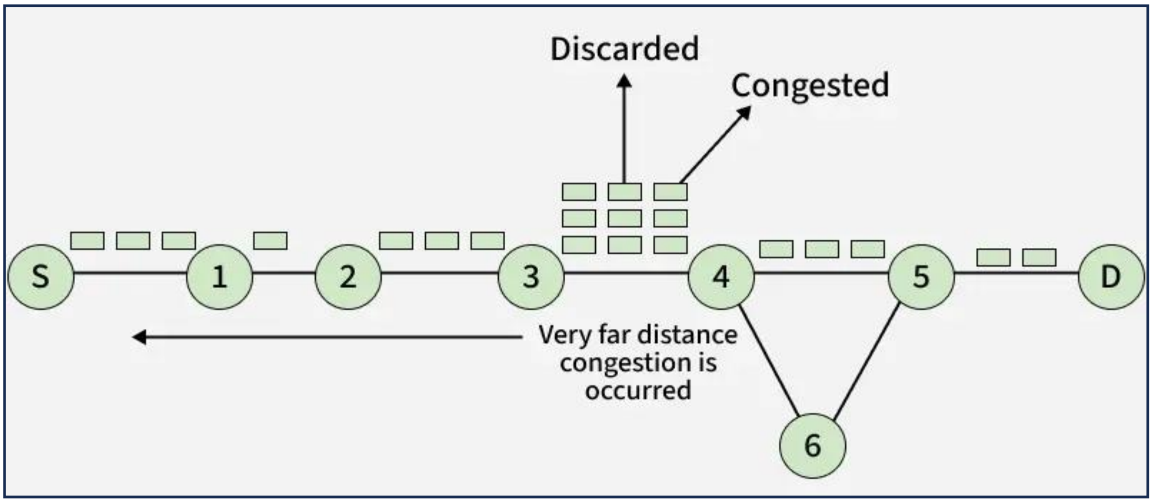
If header has error:

- Packet discarded
- ICMP Parameter Problem sent

Example

Wrong header length value.

4. Source Quench Message (Obsolete)



Previously used for congestion control.

Type

Type = 4

Working

When router became congested:

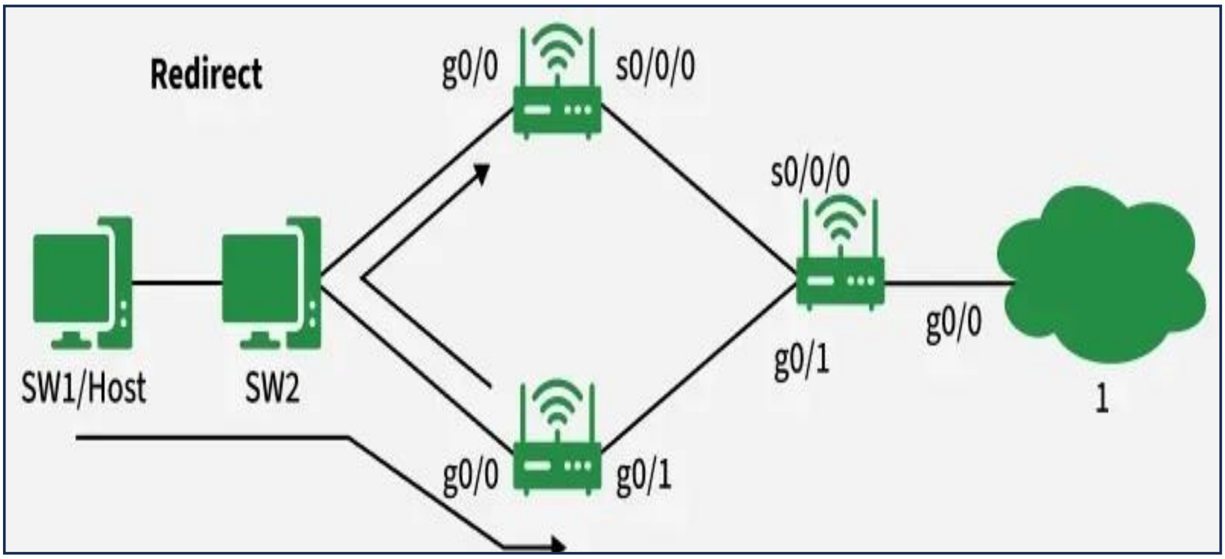
- It asked sender to slow down transmission.

Status: Obsolete now.
Modern networks use TCP flow control and congestion control instead.

Example

Router buffer becomes full due to heavy traffic.

5. Redirect Message



Informs host about a better route to destination.

Type

Type = 5

Working

Suppose:

- Host sends packet to Router A
- Router A realizes Router B is better

Router A forwards packet AND sends ICMP Redirect to sender.

Benefit

Improves routing efficiency.

Example

Host initially uses wrong gateway.

Router corrects it using redirect message.

IGMP (Internet Group Management Protocol)

Introduction

The **Internet Group Management Protocol (IGMP)** is a communication protocol used in IPv4 networks to manage multicast group memberships. It allows hosts and routers to establish and maintain multicast group communications.

IGMP enables a device to join or leave a multicast group so that multicast data can be delivered only to interested hosts.

What is Multicasting?

Multicasting is a method of communication in which data is transmitted from one sender to multiple selected receivers simultaneously.

- **Unicast:** One sender → One receiver
- **Broadcast:** One sender → All devices
- **Multicast:** One sender → A specific group of receivers

Example:
Online live streaming, video conferencing, IPTV and online gaming often use multicasting.

Purpose of IGMP

IGMP is used to:

- Manage multicast group memberships.
- Inform routers which hosts want to receive multicast traffic.
- Reduce unnecessary network traffic.
- Support efficient group communication.

Features of IGMP

- Operates at the **Network Layer**.
- Used only in **IPv4** networks.
- Supports multicast communication.
- Helps routers track multicast group members.
- Works between hosts and neighboring routers.

Working of IGMP

Step 1: Joining a Multicast Group

When a host wants to receive multicast data, it sends an **IGMP Membership Report** message to the router.

Step 2: Router Query

Routers periodically send **IGMP Query** messages to check whether hosts are still interested in receiving multicast traffic.

Step 3: Membership Report

Hosts respond with Membership Reports if they still want to remain in the multicast group.

Step 4: Leaving the Group

When a host no longer wants multicast traffic, it sends an **IGMP Leave Group** message.

Types of IGMP Messages

Message Type	**Function**
Membership Query	Sent by routers to check active group members
Membership Report	Sent by hosts to join or remain in a group
Leave Group	Sent by hosts to leave a multicast group

Versions of IGMP

1. IGMP Version 1 (IGMPv1)

- Basic multicast group management.
- No leave message support.

2. IGMP Version 2 (IGMPv2)

- Introduced Leave Group messages.
- Faster group management.

3. IGMP Version 3 (IGMPv3)

- Supports source-specific multicast (SSM).
- Allows hosts to accept multicast traffic from selected sources only.

IGMP Packet Format

An IGMP message contains:

Field	**Description**
Type	Specifies the IGMP message type
Maximum Response Time	Time allowed for response
Checksum	Error checking
Group Address	Multicast group address

Example of IGMP

Suppose several students want to watch a live online class stream:

1. Their computers join the multicast group using IGMP.
2. The router receives the membership requests.
3. The router forwards the video stream only to the interested students.
4. Students leaving the class send Leave Group messages.

This reduces unnecessary network traffic.

Advantages of IGMP

- Efficient bandwidth utilization.
- Reduces unnecessary traffic.
- Supports multimedia applications.

- Enables scalable group communication.

Limitations of IGMP

- Works only with IPv4.
- Requires multicast-enabled routers.
- Can become complex in large networks.

Difference Between IGMP and ICMP

IGMP	**ICMP**
Used for multicast group management	Used for error reporting and diagnostics
Supports multicast communication	Supports network troubleshooting
Works between hosts and routers	Works between network devices